



Scan: dataset, code,
& reproducibility info

OELLM

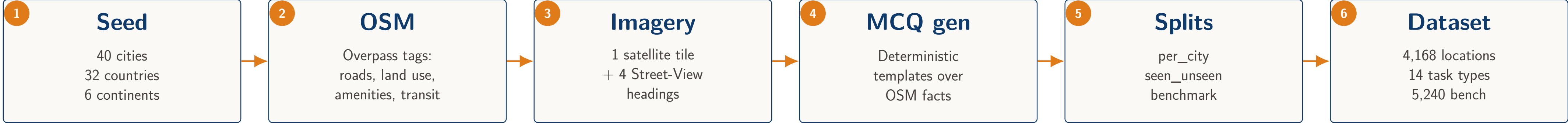
Observation of Earth · Language Model

Dataset, Model & Benchmark for Urban Understanding & Cross-View Geolocalisation

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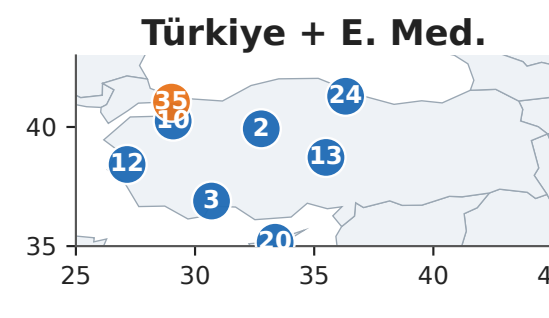
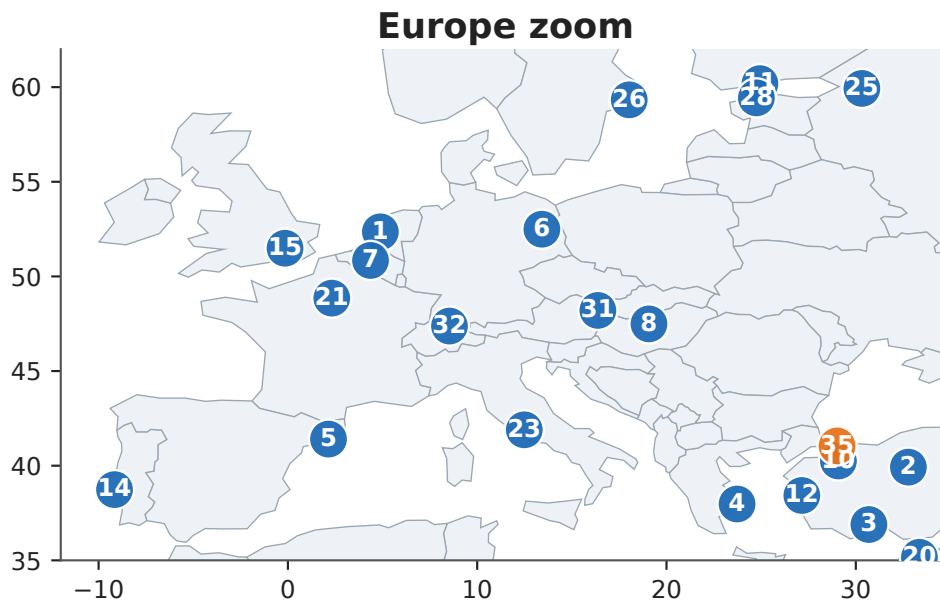
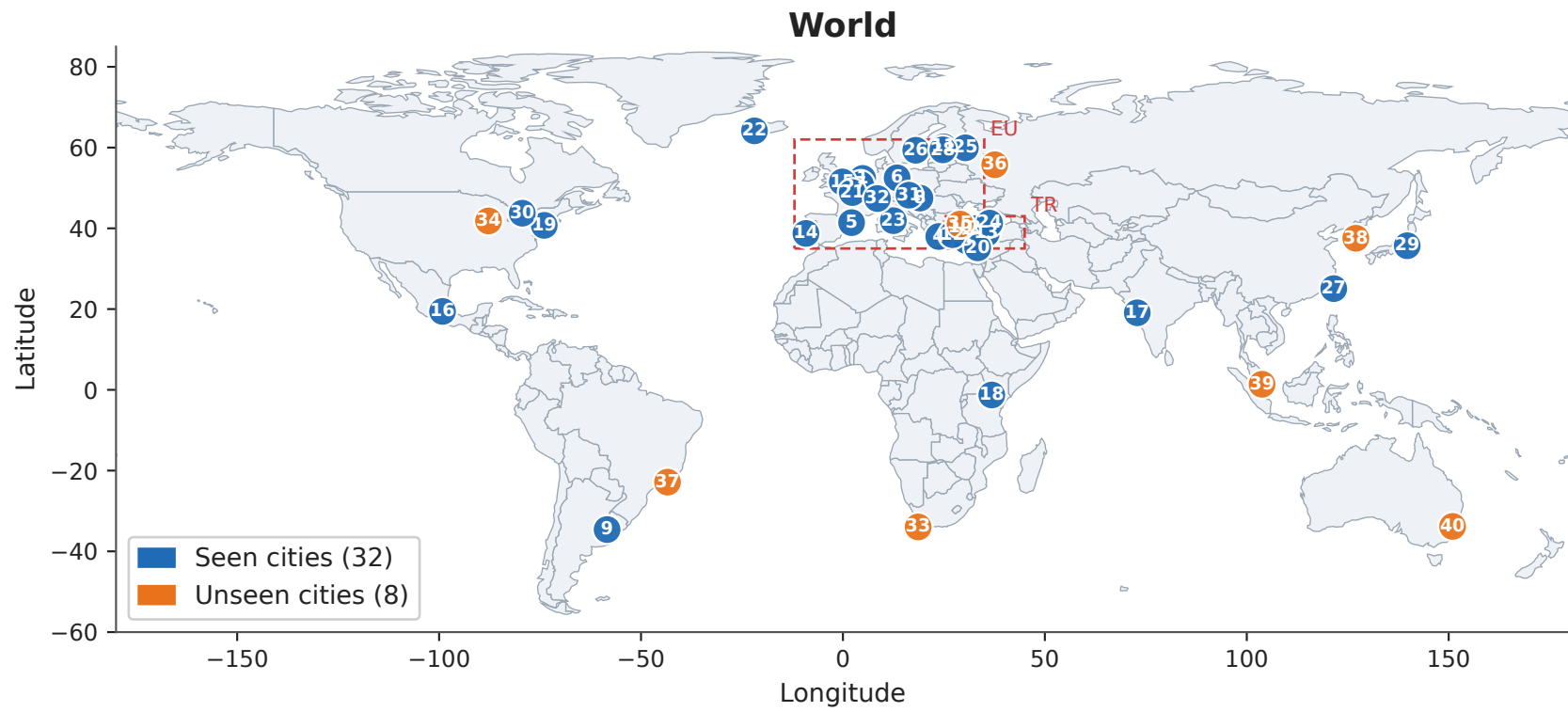
1 • The Dataset

Pipeline — OSM seeds → paired imagery → deterministic MCQs

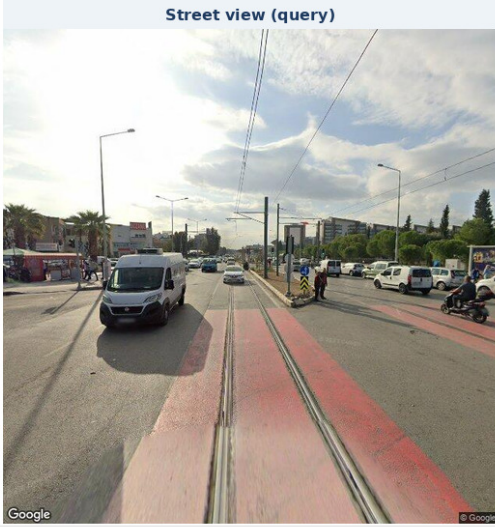
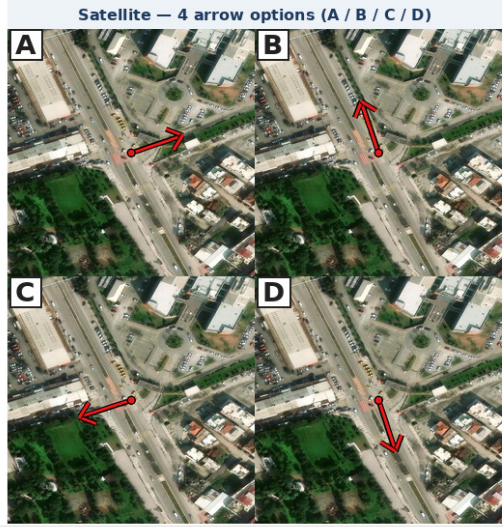


Each location pairs one 512×512 satellite tile with four Street-View headings; every answer comes **deterministically from OpenStreetMap** — no annotators, bit-exact regenerable.

Geographic coverage — 40 cities, 6 continents, 32 countries



- | | | | |
|----|---------------|----|----------------|
| 1 | Amsterdam | 21 | Paris |
| 2 | Ankara | 22 | Reykjavik |
| 3 | Antalya | 23 | Rome |
| 4 | Athens | 24 | Samsun |
| 5 | Barcelona | 25 | St. Petersburg |
| 6 | Berlin | 26 | Stockholm |
| 7 | Brussels | 27 | Taipei |
| 8 | Budapest | 28 | Tallinn |
| 9 | Buenos Aires | 29 | Tokyo |
| 10 | Bursa | 30 | Toronto |
| 11 | Izmir | 31 | Vienna |
| 12 | Helsinki | 32 | Zurich |
| 13 | Kayseri | 33 | Cape Town |
| 14 | Lisbon | 34 | Chicago |
| 15 | London | 35 | Istanbul |
| 16 | Mexico City | 36 | Moscow |
| 17 | Mumbai | 37 | Rio de Janeiro |
| 18 | Nairobi | 38 | Seoul |
| 19 | New York City | 39 | Singapore |
| 20 | Nicosia | 40 | Sydney |



camera_direction · Izmir, Turkey

Select the satellite image whose red arrow matches the viewing direction of this street view.

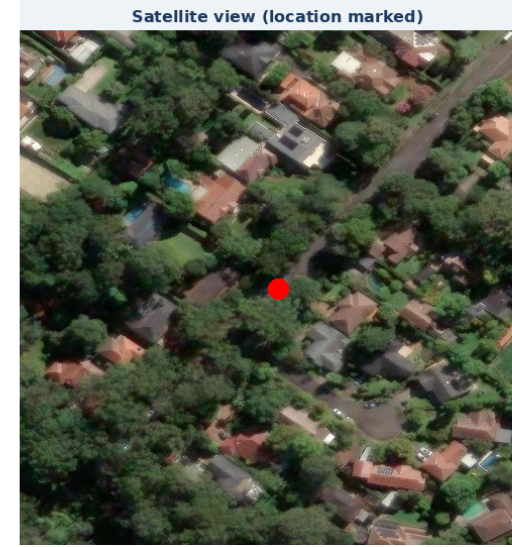
A. Image A · **B. Image B [correct]** · C. Image C · D. Image D



junction_type · Helsinki, Finland

Looking at this location, how would you classify the road junction type?

A. Signalized intersection · B. Roundabout · **C. Unsignalized intersection [correct]** · D. Grade-separated (overpass/underpass)



mismatch_mcq_easy · Sydney, Australia

Which quadrant of the street view grid matches the marked satellite location?

A. Top-Right [correct] · B. Bottom-Right · C. Top-Left · D. Bottom-Left

2 • Method & Dataset Composition

Training setup

Model Qwen3.5-4B-VL, vision SFT

Adapter LoRA $r \in \{16, 32\}$, $\alpha=r$, all-linear

Optim. bs 86–96, lr 2.5×10^{-4} , cosine, 100 warmup

Hardware RTX Pro 6000, 96 GB, single GPU

Eval greedy, deterministic, SEED = 3407

PC-base 6 ep; SU-base 8 ep; ablations 5 ep, early-stop patience 2. Every run evaluated on the full 14-topic mix.

Three splits, one taxonomy

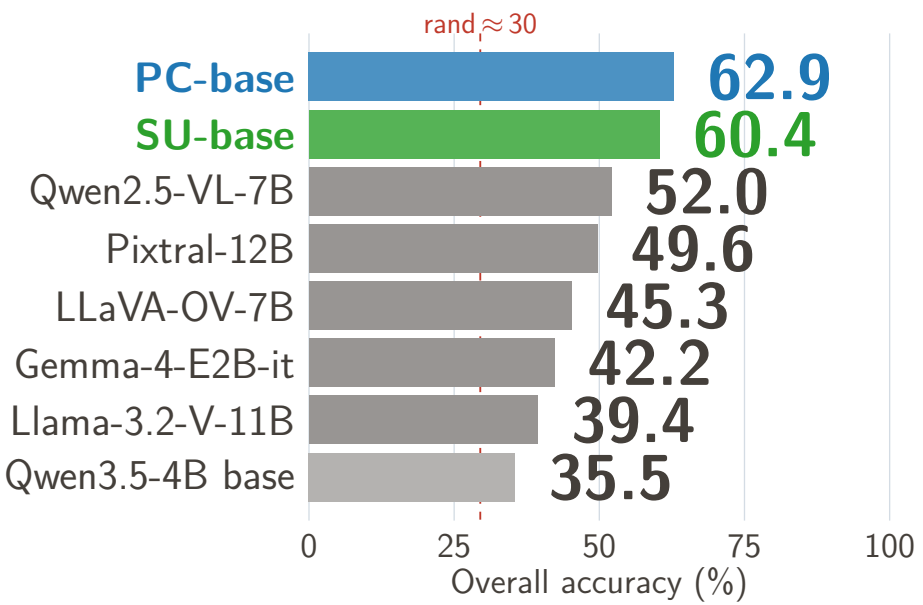
Split	Train	Val	Generalisation question
per_city	26,931	6,984 new loc.,	seen cities
seen_unseen	26,963	8,831 8 cities held out	
benchmark	5,240 pub	5,240 gold	disjoint from both

per_city = familiar contexts. seen_unseen = geographic generalisation. benchmark ships answer: null alongside a gold-labelled twin.

3 • Results — training works, but transfer has limits

Held-out benchmark — vs. 6 open VLMs

5,240 records · greedy decoding



PC-base and SU-base (ours, 4B params) beat all 6 open VLMs incl. Qwen2.5-VL-7B and Pixtral-12B. SU-base is the honest cross-city number; PC-base is best overall but overfits seen cities.

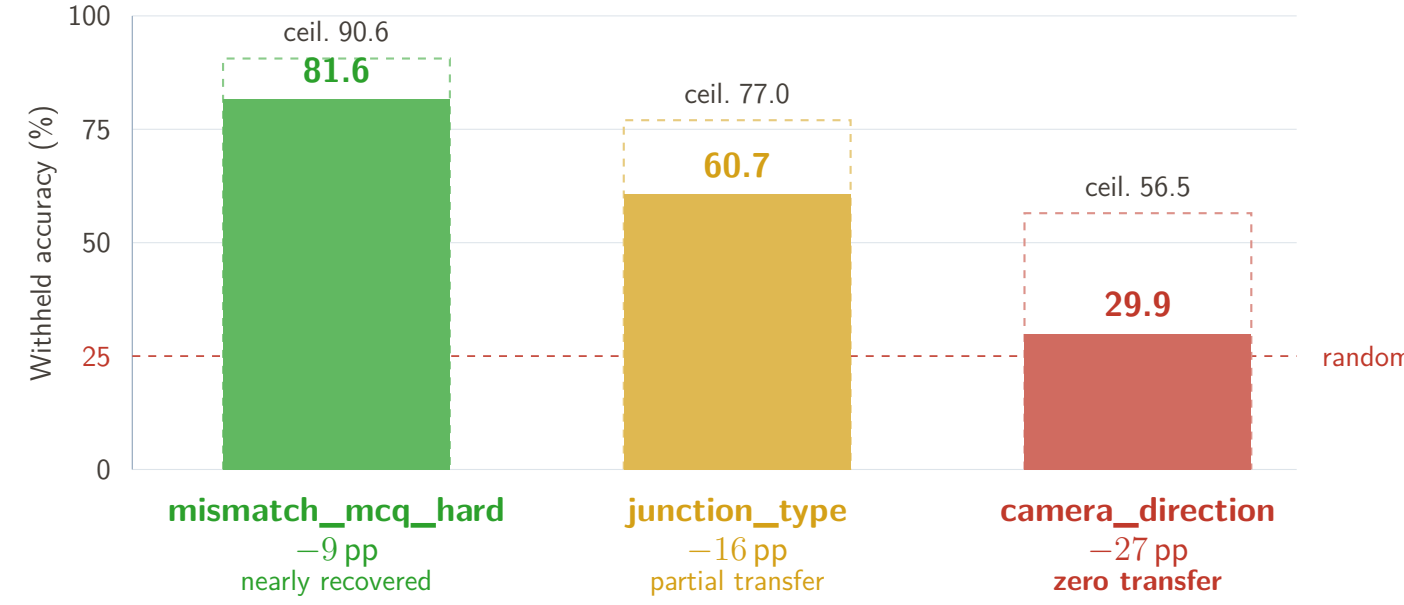
Per-task accuracy & ablations

Task	Llama 11B	LLaVA 7B	Pix 12B	Qwen2.5 7B	Qwen3.5 4B	SU ours	PC ours
Amenity Rich.	24.2	35.2	32.1	31.4	24.7	56.5	53.2
Building Height	49.0	57.9	59.0	60.5	24.7	60.5	66.3
Camera Direction	25.4	24.9	20.7	27.8	7.6	26.1	24.5
Green Space	49.8	76.9	66.0	84.7	40.9	62.6	100.0
Junction Type	44.6	53.6	49.4	61.1	40.1	70.4	75.8
Land Use	41.1	57.5	46.3	53.7	29.0	75.1	73.6
Mism. Bin. Easy	53.4	54.4	82.7	85.5	51.8	55.6	53.0
Mism. Bin. Hard	44.2	44.9	87.2	64.9	45.1	49.6	44.4
Mism. MCQ Easy	23.5	29.2	26.4	42.3	31.6	73.2	76.5
Mism. MCQ Hard	20.4	23.8	28.0	37.3	30.9	59.6	60.1
Road Surface	86.5	84.5	86.1	92.4	71.6	75.3	96.0
Road Type	45.8	59.1	58.7	54.2	45.8	73.6	70.8
Transit Density	33.7	28.7	31.8	27.3	23.5	44.7	47.7
Urban Density	34.7	39.0	44.4	40.1	37.3	69.8	71.3
Overall	39.4	45.3	49.6	52.0	35.5	60.4	62.9

Ablations (per_city: All / Trained / Withheld)

geo only (S-1)	66.3	91.4	47.1
urban+cam (S-3)	63.2	76.0	50.4
no camera (S-5)	76.5	80.1	29.9
no-hard mism. (S-6)	76.4	79.6	82.2

Transfer ladder



3 withheld-task outcomes vs ceiling (PC-base on the same topic). mismatch_mcq_hard nearly recovers via easy-variant transfer; junction_type partially transfers from geo supervision; camera_direction has no transfer source in this dataset.

Within-family transfer — easy carries hard

Withheld-task transfer (only runs where the topic is withheld are plotted)

